⊛ Prob. Statement

↳ Find Schema & tables.

(data-modelling)

Q You've mobile devices (android & iOS)...
w/ various prop.
(OS, Color, Yr. of Mfg., etc.) along w/ owner details
You need to send data [battery % & GPS]
to server

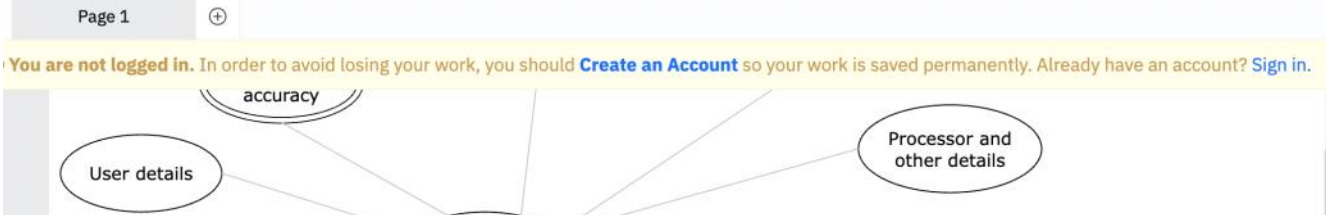
Q → 1.) Use of the data ?

↳ Hardening of new GPS sys.
(Test Reliability)
↳ Dynamic Pricing based on
battery %.
↳ Monitor Battery Health
↳ Find My Device!

∴ Decide freq. of data storage as per finalized use-case.

Note → No Arch. Diag.
↳ Just Schema — what tables,
cols,
rel^{n}-ships (if any)
⋮

Breakout (Vinod)



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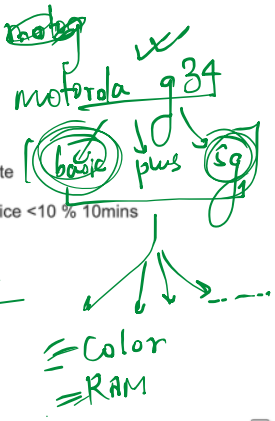
List of tables
- master/transaction

master :
1. User details - id, name, contact info, age,creation_date, update_date, device_id
2. Device details- id, type,model no, gps, os_version, mfd_date,creation_date, update_date

Transaction table
3. **performance_details**- id, user_id,batter_perc,gps_loc, creation_date - **5min**- user device <10 % 10mins

1 101 40% , gps_loc1,9:38 ist
2 101 35% gpc_loc2, 10 Pm ist
3 101 32% gpc_loc3, 10:12 Pm ist

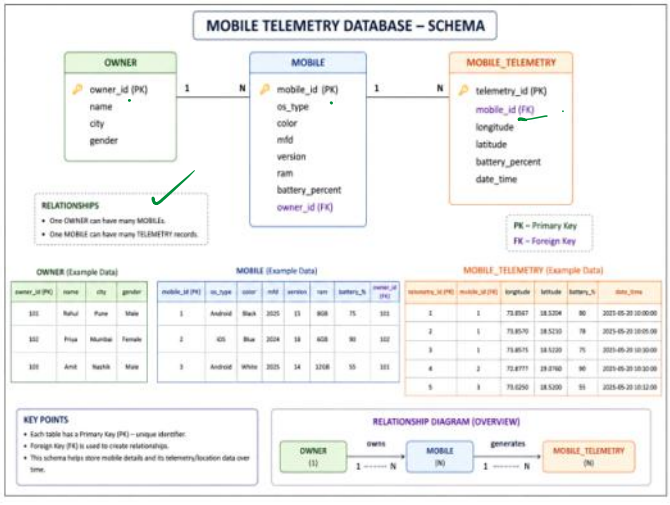
features:
1. user1- timeframe of batter drain %- group by device_type-Android, Ios
2. gps accuracy-- group by device_type-Android, Ios
3. gps - batter percentage accuracy-group by device_type-Android, Ios
4. userid- traffic-citywise



Shubhada #8

SCHEAM
name of table –
Mobile
1) ID
2)OS Type
3)color
4)mfd
5)version
6)RAM
7)BATTERY %
8)Ower_ID – F
(name ,
Name of table –
Mobile telemetry

1) ID -- F
2)longitude
3)latitude
4)battery %
5) date and time



Ower details -
1)ID--- P

2) Name
3) city
4) Gender

(*) Prob. Statement (detailed):

1. Users → buy phone
2. ↳ can have multiple phones.
3. phones → iOS / Android.
4. Phone → yr. of mfg, color, RAM, storage, processor.
5. {Monitor the health of device
   ↳ GPS, Battery %}

time-based data → Find My Device:, Track for compliance.

instant-based (last known val.) → single value. → Reliability monitoring.

(A) (B) Device + User + DeviceHealth

Hash values

8 GB + 128 GB

30 to 40 Brands → models → variants

Device → still not registered:

clubbing is fine

Model : PhoneId, Company, Model, Processor

Model : PhoneId, Company, Model, Processor
(PK)

Abstract
Model Details : (PhoneId, VariantId) Color, RAM, ...
(PK)

Device : (PhoneId, VariantId), DeviceId, IMEI
FK PK (a, b)

Users : UserId, name, email, phoneNo, ...
(PK)

(Relationship table)
UserDevice :- (UserId, DeviceId)
PK

↳ Why not userId into Device
↳ unnecessarily multiple NULLs in that col.
↳ What if one device may have multiple concurrent owners.

Device Health :- DeviceId,
GPS,
Battery Health
UpdatedAt.